HunterMills

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Seasoned Data Scientist and Machine Learning Engineer with 8+ years of end-to-end AI product delivery. Proven ability to translate challenges into scalable, production-grade solutions that integrate with medical systems, cloud platforms, and embedded applications. 15+ publications spanning rare disease prediction, bias mitigation, to deep learning on EKG.

Experience

- **Senior Machine Learning Engineer** — October 2025 - Present
  Visual Thought Labs — San Francisco, CA
  - Developed ambient drug OCR in VA operating rooms, deployed on Nvidia Jetson at 77% lower hardware cost and 73% lower latency; built the SQL pipeline populating it, and HL7 system issuing results.
- **Senior Data Scientist** — January 2022 - October 2025
  UCSF — San Francisco, CA
  - Built "Drifter," a data-drift monitoring tool used by 80 researchers across six hospitals; surfaced feature, label, and prediction drift that triggered timely ETL updates.
  - Directed a nation-wide matched-cohort study on traumatic-brain-injury outcomes, quantifying a 67% higher risk of malignant brain tumors (HR = 1.67) and publishing findings in JAMA Network Open.
- **Machine Learning Engineer** — October 2021 - January 2022
  Syntegra — San Francisco, CA
  - Created a GenAI synthetic patient data generator that transformed structured EHR records into natural-language sentences, trained BERT models, and regenerated structured data, enabling safe data sharing for rare-case research.
- **Lead Data Scientist** — December 2020 - October 2021
  LifeBell AI — Atlanta, GA (remote)
  - Deployed early sepsis detection system using XGBoost, with clinical explainer at Naval Medical Center San Diego.
  - Designed a Dynamic-Time-Warped Gaussian Mixture Model for phenotype discovery, surfacing sepsis-risk subgroups several hours before conventional detection.
- **Data Scientist** — August 2018 - December 2020
  UCSF — San Francisco, CA
  - Developed deep-learning EKG classifier for pulmonary hypertension, achieving an AUC of 0.89 and early-diagnosis capability up to two years before conventional testing.
  - Led feasibility analyses that helped secure $5M+ in research partnerships (J&J, Roche, Genentech).
- **Embedded Systems Engineer** — May 2014 - August 2018
  NASA Education & Public Outreach — Rohnert Park, CA
  - Conceived/built/launched EdgeCube 10cm satellite, overseeing sensor integration, data logging, and radio telemetry.

Education

Stanford University	MS in Computational and Mathematical Engineering	2018
Sonoma State University	BS in Physics and Mathematics	2014

Skills

Languages	Python, SQL, Julia, C, C++, JavaScript, TypeScript, LaTeX
ML Frameworks	PyTorch, TensorFlow, Scikit-Learn, XGBoost, CVXPY, RAPIDS, JAX, TensorRT, Ollama, HuggingFace, StatsModels, lifelines
Infrastructure & Compute	Spark, Spark-SQL, Hadoop, MPI, OpenMP, CUDA, AWS, GCP, Docker
Visulaiztion	Dash, Plotly, Matplotlib, Seaborn

Publications (Selected)

- C. Halabi, **H. Mills**, A. M. DiGiorgio, et al. "Civilian Traumatic Brain Injury Is Associated with Longitudinally Increased Risk of PTSD, Suicidality, and Other Mental Health Diagnoses". In: *Journal of Neurotrauma* (2026), p. 08977151251414145.
- A. S. Tang, K. P. Rankin, G. Cerono, S. Miramontes, **H. Mills**, J. Roger, et al. "Leveraging electronic health records and knowledge networks for Alzheimer's disease prediction and sex-specific biological insights". In: *Nature Aging* 4.3 (2024), pp. 379–395.
- M. A. Aras, S. Abreau, **H. Mills**, L. Radhakrishnan, et al. "Electrocardiogram detection of pulmonary hypertension using deep learning". In: *Journal of Cardiac Failure* 29.7 (2023), pp. 1017–1028.